

Bachelor in Computer Applications

Course Title: **Data Structure and Algorithms**

Code No: Semester: **3**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

1. What is stack? Explain different stack operations. Explain algorithm to evaluate postfix expression.
2. Explain almost complete binary tree with example. How do you insert, search, and delete nodes in a binary search tree? Explain with suitable example?
3. Discuss the limitation of choosing first element as pivot in quick sort. Using merge sort algorithm, sort the numbers 40, 6, 5, 21, 3, 100, 90, 7, 8, 12, 30.
4. Define data type and ADT. What are the benefits of using ADT? Explain
5. What is space complexity? Explain omega notation with example.
6. What is circular queue? How can you implement circular queue?
7. Define recursion. Explain Tower of Hanoi (TOH) with example.
8. How can we use linked list to implement queue? Explain.
9. What are different applications of binary tree? Explain.
10. Why do we need hashing? Explain quadratic probing.
11. Define spanning tree. Explain minimum spanning tree with example.
12. Write short notes on:
a. Doubly circular linked list
b. Breadth first search